

(VM) sub-system from the operating system? If you cannot completely eliminate the VM sub-system, what are the changes in paging algorithms (both data structures and page replacement algorithms) that you would like to implement so that the cost of the virtual memory sub-system is minimised for this particular server?

3. You were asked to create a binary classification system using the supervised learning approach, to analyse X-ray images when detecting lung cancer. You have implemented two different approaches and would like to evaluate both to see which performs better. You find that Approach 1 has an accuracy of 97 per cent and Approach 2 has an accuracy of 98 per cent. Given that, would you choose Approach 2 over Approach 1? Would accuracy alone be a sufficient criterion in your selection? (*Clue:* Note that the cancer image data set is typically imbalanced – most of the images are non-cancerous and only a few would be cancerous.)
4. Now consider Question 3 again. Instead of getting accuracies of 98 per cent and 97, both your approaches have a pretty low accuracy of 59 per cent and 65 per cent. While you are debating whether to implement a totally different approach, your friend suggests that you use an ensemble classifier wherein you can combine the two classifiers you have already built and generate a prediction. Would the combined ensemble classifier be better in performance than either of the classifiers? If yes, explain why. If not, explain why it may perform poorly compared to either of the two classifiers. When is an ensemble classifier a good choice to try out when individual classifiers are not able to reach the desired accuracy?
5. You have built a classification model for email spam such that you are able to reach a training error which is very close to zero. Your classification model was a deep neural network with five hidden layers. However, you find that the validation error is not small but around 25. What is the problem with your model?
6. Regularisation techniques are widely used to avoid overfitting of the model parameters to training data so that the constructed network can still generalise well for unseen test data. Ridge regression and Lasso regression are two of the popular regularisation techniques. When would you choose to use ridge regression and when would you use Lasso? Specifically, you are given a data set, in which you know that out of 100 features, only 10 to 15 of them are significant for prediction, based on your domain knowledge. In that case, would you choose ridge regression over Lasso?
7. Now let us move away from machine learning and data science to computer architecture. You are given a computer system with 16 cores, 64GB RAM, 512KB per core L1 cache (both instruction and data caches are separate and each is of size 512KB), 2MB per core L2 cache, and 1GB L3 cache which is shared among all cores. L1 cache is direct-mapped whereas L2 and L3 caches are four-way set associative. Now you are told to run multiple MySQL server instances on this machine, so that you can use this machine as a common back-end system for multiple Web applications. Can you characterise the different types of cache misses that would be encountered by the MySQL instances?
8. Now consider Question 7 above, where you are not given the cache sizes of L1, L2 and L3, but have been asked to figure out the approximate sizes of L1, L2 and L3 caches. You are given the average cache access times (hit and miss penalty for each access) for L1, L2 and L3 caches. Other system parameters are the same. Can you write a small code snippet which can determine the approximate sizes of L1, L2 and L3 caches? Now, is it possible to figure out the cache sizes if you are not aware of the average cache access times?
9. Consider that you have written two different versions of a multi-threaded application. One version extensively uses linked lists and the second version uses mainly arrays. You know that the memory accesses performed by your application do not exhibit good locality due to their inherent nature. You also know that your memory accesses are around 75 per cent reads. Now you have the choice between two computer systems — one with an L3 cache size of 24MB shared across all eight cores, and another with an L3 cache size of 2MB independent for each of the eight cores. Are there any specific reasons why you would choose System 1 rather than System 2 for each of your application versions?
10. Can you explain the stochastic gradient descent algorithm? When would you prefer to use the co-ordinate gradient descent algorithm instead of the regular stochastic gradient algorithm? Here's a related question. You are told that the cost function you are trying to optimise is non-convex. Would you still be able to use the co-ordinate descent algorithm?

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, wishing all our readers a wonderful and productive month ahead! 

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